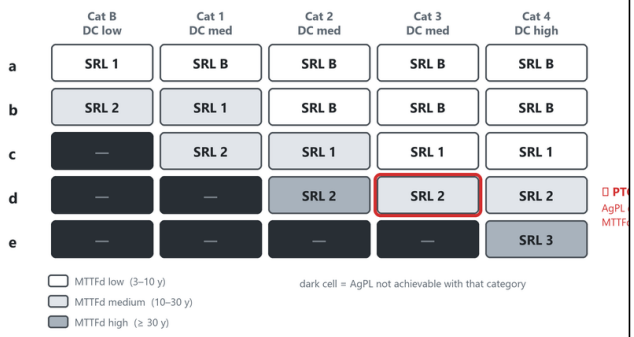
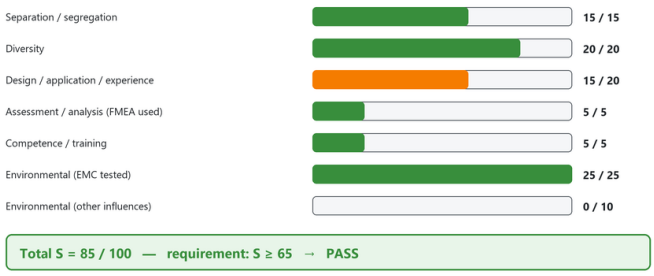
Functional_Safety_Concept

ReqID	Contents	AgPL	Allocation	Derived from	Status
PTO-FSC-000	**Functional Safety Concept — Rear PTO** Safety goals and functional safety requirements per ISO 25119-2:2018, Clause 7. Every hazardous situation with AgPLr > QM maps to a safety goal; FSRs inherit the AgPLr of their goal. The selected combination of category, MTTFd, DC and SRL that achieves each AgPL is recorded in §3.	QM	System	—	Verified
—	**1. Safety goals**	QM	System	—	Verified
PTO-SG-01	The PTO shall not engage unless a deliberate, valid operator engage command is present. Safe state: clutch valve de-energised → clutch open, spring brake applied.	AgPL d	System	PTO-HZ-01, -02	Verified
PTO-SG-02	On a stop command or a stop condition the PTO shall be disengaged within 400 ms; the stop function has priority over any engage request (Annex F.3).	AgPL c	System	PTO-HZ-03	Verified
PTO-SG-03	The PTO shall not re-engage automatically: after power-up, after engine stall/restart, and after operator-presence loss, re-engagement requires a new deliberate command from the OFF position (manual reset, Annex F.4/F.5).	AgPL c	System	PTO-HZ-04, -05	Verified
PTO-SG-04	External (fender) engagement shall act only while the button is actively held, shall limit the shaft to creep speed (≤ 20 rpm), and shall never latch.	AgPL b	System	PTO-HZ-06	Verified
PTO-SG-05	The PTO shall keep the shaft inside the selected mode's speed envelope (≤ +10 %) and shall transfer torque through a controlled engagement ramp.	AgPL b	System	PTO-HZ-07, -08	Verified
—	**2. Functional safety requirements**	QM	System	—	Verified
PTO-FSR-001	An engage command shall be accepted only as the valid two-step cab sequence (collar lifted, then switch pressed) with both switch contacts agreeing within 50 ms.	AgPL d	Sensor / input	PTO-SG-01	Verified
PTO-FSR-002	Engagement shall require independent agreement of two processing channels; no single E/E/PES fault shall energise the clutch valve.	AgPL d	ECU / Hardware	PTO-SG-01	Verified
PTO-FSR-003	At power-up the PTO output shall stay disabled until the start interlock passes: self-test complete, engage command absent, presence state valid (Annex F.2).	AgPL d	ECU / Software	PTO-SG-01, -03	Verified
PTO-FSR-004	A detected dangerous fault in the engage path shall force and latch the safe state; recovery only via manual reset with the fault cleared.	AgPL d	System	PTO-SG-01	Verified
PTO-FSR-005	A stop demand (switch OFF, either contact opening, presence loss > 1 s in mobile mode, monitor request) shall always be executable independently by each channel.	AgPL c	ECU / Hardware	PTO-SG-02	Verified
PTO-FSR-006	The reaction chain stop-demand → valve de-energised → clutch open → brake applied shall complete within 400 ms (response time per 25119-1, 3.38).	AgPL c	System	PTO-SG-02	Verified
PTO-FSR-007	The engaged state shall not be restored from memory: no engage state may survive a power cycle, brown-out or ECU reset.	AgPL c	ECU / Software	PTO-SG-03	Verified
PTO-FSR-008	Operator presence shall be monitored with plausibility (two contacts); in mobile operation, presence loss > 1 s shall trigger the stop chain. Stationary work mode requires a deliberate enable and engine-speed limit.	AgPL c	Sensor / input	PTO-SG-03	Verified
PTO-FSR-009	Fender engagement shall be accepted only with the cab switch OFF, shall run the shaft at creep (≤ 20 rpm, jog pulses), and shall stop within 200 ms of button release.	AgPL b	System	PTO-SG-04	Verified
PTO-FSR-010	Fender-button stuck-at conditions shall be detected (release supervision) and shall inhibit external engagement.	AgPL b	Sensor / input	PTO-SG-04	Verified
PTO-FSR-011	Shaft speed shall be supervised against the selected mode; a sustained excursion beyond +10 % shall raise a warning and trigger a controlled disengage (degraded stop).	AgPL b	ECU / Software	PTO-SG-05	Verified
PTO-FSR-012	The selected speed mode shall be read redundantly (coded lever + measured engine/shaft ratio) and mismatches shall inhibit engagement.	AgPL b	Sensor / input	PTO-SG-05	Verified
PTO-FSR-013	Engagement shall modulate clutch pressure along a ramp (≤ 2.0 s, slip supervised); step engagement shall be prevented.	AgPL b	ECU / Software	PTO-SG-05	Verified
PTO-FSR-014	Detected faults and degraded modes shall be indicated to the operator (lamp + buzzer via cluster); indication is QM and shall not be relied on for any safety function.	QM	System	PTO-SG-02	Verified
PTO-FSC-SEL	**3. Achieving the AgPL — selected combination (25119-2, 7.3.5)** **Achieving an AgPL (ISO 25119-2, Figure 2) — cell = required SRL, shade = required MTTFd**  *Figure 1 — category/MTTFd/DC/SRL combinations; the PTO-ECU selection for AgPL d is ringed* Selected for the AgPL d engage/stop chain: Category 3 (dual channel, cross-monitoring), MTTFd medium per channel (target ≥ 10 y, achieved values in 4_Hardware/Reliability_Metrics), DC medium (target ≥ 60 %, achieved by the diagnostic set D1–D5), SRL 2 software, CCF score ≥ 65 (achieved 85, Annex D scoring below). AgPL b/c functions ride on the same architecture and inherit its integrity. **Common-cause failure scoring (ISO 25119-2 Annex D) — PTO-ECU dual channel**  *Figure 2 — CCF scoring (Annex D)*	AgPL d	System	—	Verified
PTO-FSC-EXT	**4. External measures and interfaces to other technologies** The spring-applied PTO brake and the clutch return spring are mechanical measures: de-energising the valve reaches the safe state without any E/E/PES action (de-energise-to-safe). The master shield and shaft guards are machine-level measures per ISO 4254-1 (outside the UoO). Failures of hydraulic and mechanical parts are context events in the FTA, not SRP/CS failures (25119-1, 6.2.2).	QM	System	—	Verified